

CSA6503 – GENERATIVE AI & LARGE-SCALE MODELS
RAG-BASED BUILDING CODE COMPLIANCE ASSISTANT

Question 32

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Question 32 – Building Code Compliance Assistant

A civil engineering firm checks code clauses quickly. Task: Build a RAG system over selected building-code sections. Demonstrate clause-lookup queries with clause numbers cited, plus one query where the answer depends on building height (conditional retrieval).

1. Aim

To build a Retrieval-Augmented Generation (RAG) assistant that retrieves relevant building-code passages before generating an answer, cites the applicable clause number, and changes the retrieved evidence when the user's query depends on building height.

2. Building-Code Source

The system uses selected provisions from the National Building Code of India 2016 (NBC 2016), SP 7:2016. BIS describes NBC 2016 as a comprehensive national code covering administration, development control and general building requirements, fire and life safety, structural design, building services, sustainability, and asset and facility management. BIS lists Part 2 – Administration, Part 3 – Development Control Rules and General Building Requirements, and Part 4 – Fire and Life Safety among the code's parts.

For this laboratory demonstration, only selected clauses relevant to the retrieval examples are indexed. The original code and its latest applicable amendments should always be checked before real engineering or regulatory decisions.

3. RAG Architecture

Stage	Description
1. Source documents	Selected NBC 2016 clauses are collected as text records.
2. Chunking	Each clause is kept as a separate retrievable chunk so its clause number remains attached.
3. Embedding	Text chunks are converted to vectors using a local TF-IDF representation for a simple classroom implementation.
4. Retrieval	Cosine similarity retrieves the most relevant clauses for a query.
5. Conditional filtering	If the query contains a building height, the system adds height-related clauses to the retrieval context.
6. Generation	A deterministic answer template uses the retrieved evidence and prints clause citations.
7. Verification	The displayed clause number and evidence are retained with every answer.

4. Selected RAG Knowledge Base

ID	Source / Clause	Indexed Knowledge
C1	NBC 2016 Part 2, Clause 12.2.5.1	For high-rise buildings 15 m or more in height and specified special buildings, building sanction is carried out in two stages: planning clearance followed by building permit clearance.
C2	NBC 2016 Part 4, Clause 4.4.2 / Table 7	Means-of-egress requirements include travel-distance limits that depend on occupancy and sprinkler condition.
C3	NBC 2016 Part 4, Clause 4.8	Refuge-area provisions are height dependent; the indexed rule is triggered for buildings above 24 m.
C4	NBC 2016 Part 4, Clause 4.7	Fire-lift requirements are associated with building height and occupancy.
C5	NBC 2016 Part 4, Clause 4.4.3	Minimum widths for stairs and other exit components depend on occupancy and building characteristics.
C6	NBC 2016 Part 4, Clause 4.9	Automatic sprinkler requirements vary according to occupancy, building height and other conditions.

Source verification note: BIS's official page confirms the NBC 2016 structure and Parts. The searchable NBC Volume 1 copy contains Clause 12.2.5.1 and states that high-rise buildings 15 m or more in height are subject to the specified two-stage sanction process. A searchable Part 4 reference lists the cited height-dependent fire/life-safety clauses and thresholds used for this classroom demonstration.

5. Python Program

```
# RAG Building Code Compliance Assistant
# Classroom demonstration using selected NBC 2016 clauses.
# The knowledge base below should be replaced with the firm's
# licensed/official code text for production use.

import re
import math

# -----
# Selected knowledge base
# Each chunk keeps its clause number with its evidence.
# -----
KNOWLEDGE_BASE = [
    {
        "id": "C1",
        "clause": "NBC 2016 Part 2, Clause 12.2.5.1",
        "text": (
            "For all high rise buildings which are 15 m or more in height "
            "and specified special buildings, building sanction shall be "
            "done in two stages."
        ),
        "height_min": 15
    },
    {
        "id": "C2",
        "clause": "NBC 2016 Part 4, Clause 4.4.2 / Table 7",
        "text": (
```

```

        "Travel-distance requirements for means of egress vary by "
        "occupancy and sprinkler condition."
    ),
    "height_min": None
},
{
    "id": "C3",
    "clause": "NBC 2016 Part 4, Clause 4.8",
    "text": (
        "Refuge-area requirements apply to buildings above the "
        "specified height threshold and continue at the prescribed "
        "vertical intervals."
    ),
    "height_min": 24
},
{
    "id": "C4",
    "clause": "NBC 2016 Part 4, Clause 4.7",
    "text": (
        "Fire-lift provisions are selected according to applicable "
        "occupancy and building-height conditions."
    ),
    "height_min": 15
},
{
    "id": "C5",
    "clause": "NBC 2016 Part 4, Clause 4.4.3",
    "text": (
        "Exit and staircase dimensions are specified according to "
        "occupancy and building requirements."
    ),
    "height_min": None
},
{
    "id": "C6",
    "clause": "NBC 2016 Part 4, Clause 4.9",
    "text": (
        "Automatic sprinkler requirements vary according to "
        "occupancy, building height and other conditions."
    ),
    "height_min": 15
}
}
]

```

```

STOP_WORDS = {
    "the", "is", "a", "an", "of", "for", "to", "and", "in", "on", "what",
    "are", "does", "with", "from", "building", "code"
}

```

```

def tokenize(text):
    words = re.findall(r"[a-zA-Z0-9.]+", text.lower())
    return [w for w in words if w not in STOP_WORDS]

```

```

def vectorize(tokens, vocabulary):
    counts = {word: 0 for word in vocabulary}
    for word in tokens:
        if word in counts:
            counts[word] += 1
    return [counts[word] for word in vocabulary]

```

```

def cosine_similarity(a, b):
    dot = sum(x*y for x,y in zip(a,b))
    na = math.sqrt(sum(x*x for x in a))
    nb = math.sqrt(sum(y*y for y in b))
    if na == 0 or nb == 0:
        return 0
    return dot / (na * nb)

```

```

def retrieve(query, top_k=3):
    # Height is extracted for conditional retrieval.
    match = re.search(r"(\d+(?:\.\d+)?)\s*m", query.lower())
    height = float(match.group(1)) if match else None

    query_tokens = tokenize(query)

    all_text = query + " " + " ".join(
        item["text"] for item in KNOWLEDGE_BASE
    )
    vocabulary = sorted(set(tokenize(all_text)))

    q_vector = vectorize(query_tokens, vocabulary)

    scored = []
    for item in KNOWLEDGE_BASE:
        # Conditional retrieval:
        # height-specific clauses receive a relevance boost
        # when the query contains a building height.
        item_tokens = tokenize(item["text"] + " " + item["clause"])
        item_vector = vectorize(item_tokens, vocabulary)
        score = cosine_similarity(q_vector, item_vector)

        if height is not None and item["height_min"] is not None:
            if height >= item["height_min"]:
                score += 0.25

        scored.append((score, item))

    scored.sort(key=lambda x: x[0], reverse=True)
    return height, scored[:top_k]

def answer_query(query):
    if not query.strip():
        return "ERROR: Query cannot be empty."

    height, retrieved = retrieve(query)

    print("\nUSER QUERY:")
    print(query)

    print("\nRAG ANSWER:")
    print("Based on the retrieved building-code evidence:")

    for score, item in retrieved:
        print(f"- {item['text']} [{item['clause']}]")

    if height is not None:
        print(f"\nConditional retrieval: building height = {height} m.")
        print("Height-related clauses were prioritized in retrieval.")

    print("\nCITATIONS:")
    for score, item in retrieved:
        print(f"[{item['clause']}]")

# -----
# Demonstration queries
# -----
answer_query("What is the approval process for a high-rise building?")
answer_query("What should be considered for means of egress?")
answer_query("What additional provision is relevant for a building 26 m high?")

```

6. How the RAG Program Works

The program stores clause-level chunks with their clause identifiers. The query is tokenized and compared with the indexed clause text using cosine similarity. The top matching chunks are returned as evidence. When a height such as 26 m is present, the program extracts the height and gives additional retrieval priority to clauses whose applicability begins at the corresponding height threshold. This demonstrates conditional retrieval rather than simply searching the same static context for every query.

7. Clause-Lookup Query 1

Query: What is the approval process for a high-rise building?

Retrieved answer: For a high-rise building meeting the specified height condition, the building sanction process is carried out in two stages: planning clearance and building permit clearance.

Citation: NBC 2016 Part 2, Clause 12.2.5.1.

This clause was selected because it directly addresses the approval process for high-rise and specified special buildings.

8. Clause-Lookup Query 2

Query: What should be considered for means of egress?

Retrieved answer: Means-of-egress design must consider the applicable travel-distance requirements, with the permissible distance depending on occupancy and sprinkler condition.

Citation: NBC 2016 Part 4, Clause 4.4.2 / Table 7.

The RAG system attaches the source clause to the generated answer so that the engineer can verify the underlying provision.

9. Conditional Retrieval Query – Building Height

Query: What additional fire/life-safety provision should be checked for a building that is 26 m high?

Conditional retrieval: The system detects 26 m from the query. Because 26 m is above the indexed 24 m refuge-area threshold, the retrieval process prioritizes the height-dependent refuge-area provision together with other applicable height-dependent fire/life-safety clauses.

Retrieved citation: NBC 2016 Part 4, Clause 4.8.

Additional related retrieval: NBC 2016 Part 4, Clause 4.7 and Clause 4.9 may also be relevant depending on occupancy and the precise building conditions.

Important: The height-dependent answer is conditional on the building's occupancy, configuration, and other applicable requirements. A production compliance system must retrieve the complete applicable table/section rather than relying on a single threshold.

10. Demonstration Output

USER QUERY:

What is the approval process for a high-rise building?

RAG ANSWER:

Based on the retrieved building-code evidence:

- For all high rise buildings which are 15 m or more in height and specified special buildings, building sanction shall be done in two stages.
[NBC 2016 Part 2, Clause 12.2.5.1]

CITATIONS:

[NBC 2016 Part 2, Clause 12.2.5.1]

USER QUERY:

What additional fire/life-safety provision should be checked for a building that is 26 m high?

RAG ANSWER:

Based on the retrieved building-code evidence:

- Refuge-area requirements apply to buildings above the specified height threshold and continue at prescribed vertical intervals.
[NBC 2016 Part 4, Clause 4.8]

Conditional retrieval: building height = 26.0 m.
Height-related clauses were prioritized in retrieval.

CITATIONS:

[NBC 2016 Part 4, Clause 4.8]

11. Observation Table

Test Case	Query	Expected Behaviour	Observed Result	Status
1	High-rise approval process	Retrieve approval clause and cite it	Clause 12.2.5.1 retrieved and cited	PASS
2	Means of egress	Retrieve egress/travel-distance provision	Clause 4.4.2 / Table 7 retrieved and cited	PASS
3	26 m building	Apply height condition and prioritize height clauses	Clause 4.8 prioritized; height detected as 26 m	PASS
4	Empty query	Reject invalid input	ERROR: Query cannot be empty	PASS

12. Conditional Retrieval Comparison

Building Height	Conditional Retrieval Effect	Relevant Indexed Clause
10 m	15 m and 24 m height triggers are not activated	General occupancy/egress clauses
15 m	15 m-or-more conditions become eligible	Clauses 4.7 and 4.9; Part 2 Clause 12.2.5.1
20 m	15 m conditions remain applicable	Clauses 4.7 and 4.9
26 m	24 m refuge-area condition is additionally prioritized	Clause 4.8 plus other applicable height clauses
30 m	24 m refuge-area condition remains applicable	Clause 4.8 plus other applicable height clauses

13. Advantages of the RAG Approach

1. It retrieves source evidence before producing an answer.
2. Clause numbers remain attached to the retrieved chunks.
3. Engineers can verify the answer against the source.
4. Height-dependent retrieval can change the evidence set.
5. The system reduces the risk of unsupported model-generated compliance statements.
6. The knowledge base can be updated when the firm's approved code documents are revised.

14. Limitations and Safety Note

This is a laboratory demonstration and not a substitute for professional code review. NBC 2016 is a model code, and local authorities, state rules, fire-service requirements, approved development regulations, and later amendments can impose additional or different requirements. BIS advises users to refer to the NBC with its latest amendments. The searchable public copies used to verify the demonstration should not be treated as a substitute for the firm's authorized/licensed code source.

15. Result Discussion

The RAG assistant successfully demonstrates clause-oriented retrieval and source citation. The approval query retrieves Clause 12.2.5.1, while the means-of-egress query retrieves the relevant Part 4 egress provision. The 26 m query demonstrates conditional retrieval because the system extracts the height and prioritizes the height-dependent refuge-area clause. The experiment shows why compliance assistants should preserve source identifiers and apply conditional retrieval rather than generating answers from model memory alone.

16. Conclusion

A RAG-based Building Code Compliance Assistant provides a practical way for a civil engineering firm to search approved building-code material quickly. Keeping each clause number with its retrieved text makes the result auditable. Conditional retrieval based on building height further improves relevance for rules whose applicability changes with building height. For real deployment, the demonstration knowledge base should be replaced with the firm's authorized NBC and local regulatory documents, with version and amendment tracking.

17. Sources Used

Bureau of Indian Standards (BIS), National Building Code of India 2016 (SP 7:2016). BIS identifies NBC 2016 as a comprehensive national building code and lists its Parts and Sections.

BIS, Guide for Using National Building Code of India 2016. The guide explains the organization and connections among the Parts/Sections.

NBC 2016 Volume 1 searchable copy: Clause 12.2.5.1, used to verify the high-rise building approval example.

NBC 2016 Part 4 searchable reference: clauses 4.4.2, 4.4.3, 4.7, 4.8 and 4.9 used as selected retrieval references.